Cross-Sectional Sector Momentum in Fullgoal OTC Funds

Research summary · Personal project · Python / pandas / NumPy · github.com/auml168/FuGuoCup-quantsys

Research question

Can cross-sectional sector momentum, combined with a trend filter, generate a stable risk-adjusted return in Fullgoal OTC funds after subscription/redemption fees, confirmation and settlement lags, and purchase limits?

DESIGN AT A GLANCE

Sample	2021-01-04 → 2026-10-16
Universe	917 catalog records → 577 supported → 570 collected → 500 researched
Strategies	9 fixed strategies; 252-day initial train, 8 × 126-day rolling folds, 126-day holdout
Execution	T-2 signal · T+1 confirmation · T+3 redemption settlement
Costs	Subscription fee, FIFO redemption tiers, minimum holding, purchase limits
Result	**Only sector_rotation is positive in both rolling validation and holdout; no deployment approval**

SIGNAL

- **Factors:** 20/60/126-day momentum, 20/60-day volatility, 120-day MA deviation, 60-day drawdown, risk-adjusted momentum.
- **Sector rotation:** rank sectors by average 20-day momentum; take top 2 positive sectors; within each, require mom60 > 0 and select at most 3 funds by composite score.
- **Composite score:** 30% rank(mom20) + 30% rank(mom60) + 20% rank(risk-adjusted) + 20% rank(-vol60).
- **Constraints:** single fund ≤35%, single sector ≤50%, total exposure ≤95%, one share class per group, remainder in cash.

VALIDATION AND RESULTS

252 initial observations, eight non-overlapping 126-day rolling folds, and a 126-day holdout (2026-04-10 to 2026-10-16). Parameters are fixed; there is no parameter search and no selection based on the holdout.

Strategy	Rolling validation	Holdout	Max DD	Sharpe	Turnover	Fees (CNY)
sector_rotation	+0.21%	**+1.20%**	21.44%	0.22	9.21	6,329.89
timing	+21.18%	-6.13%	18.70%	-0.60	8.76	5,690.10
risk_budget	-6.96%	+0.37%	10.23%	0.12	4.08	2,671.70
equal_weight	Not verifiable	-7.87%	14.04%	-0.89	1.21	467.79
cash	0.00%	0.00%	0.00%	0.00	0.00	0.00

FINDINGS

- Only **sector_rotation** is positive in both rolling validation and the holdout. **timing** has the highest rolling validation (+21.18%) but a negative holdout (-6.13%), indicating regime dependence.
- The +1.20% holdout return is net of fees. Fees are ¥6,329.89 (about 3.16% of initial capital) with 9.21x turnover; costs are a binding constraint.
- The equal-weight incumbent is not reliably priceable: a holding crossed an unexplained NAV break on 2022-04-06. The segment is left empty rather than stitched into a return.
- No strategy cleared the deployment gate. The system remains research-only and does not select winners on the holdout. The holdout has already been viewed and is not a forward sample.

RESEARCH TAKEAWAY

Sector momentum is not a stable alpha in this universe: the sign is inconsistent across rolling validation and the holdout, and costs bind. The more valuable output is the research infrastructure — reproducible quality-break quarantine, an explicit OTC settlement ledger, and an honest negative result.

NEXT EXPERIMENTS

- Point-in-time fund universe and delistings; NAV release timestamps instead of the T-2 proxy.
- Robustness to sector definition and momentum window; decompose sector-selection and fund-selection alpha.
- Daily vs weekly rebalancing net of costs; purchase-limit, settlement, and capacity-aware execution.
- Forward records after freezing the design; multi-window significance tests rather than single-window returns.

Code, methodology, and reproducible evidence: https://github.com/auml168/FuGuoCup-quantsys · Personal research only; not investment advice.